claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ad92f805beea42583.jsonl`
- SHA-256: `739e7018044460e5e87a6a870b671045bb402d54b6c6df2d10ad1dcbd1bc0bcd`
- Source modified: `2026-06-16T03:19:22+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:17:48.907Z)

Web Searcher: Knowledge-distillation licensing boundary + label-noise from teacher

```
Research question: "Research question: **What concrete, recent (2022?2025) techniques and
evaluation practices should a local, commercially-licensed badminton rally-detector adopt to
close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16
golden videos)?** Produce an adversarially-verified, cited report organized around the four
dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over
generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a
heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff"
confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits
far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it
does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active"
iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold;
active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary
split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves +
knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-
frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom
licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM
(Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference;
minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture
skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) +
late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet,
See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation,
tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-
one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-
count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE,
a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment
leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal
windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-
window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ?
what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented
actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression
and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression),
BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads),
smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,
```

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Your angle: **Knowledge-distillation licensing boundary + label-noise from teacher** ? Dimension 3 legal/governance crux + Dimension 4 label noise: the boundary between using a paid teacher (Gemini) at inference for pseudo-labels vs. its outputs becoming training data, plus noisy-label handling ? the hard license constraint.
Search query: `knowledge distillation teacher student license terms training on LLM outputs prohibited learning with noisy labels weak supervision distillation 2023 2024`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.
Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.
Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-16T03:17:51.556Z)

I'll research the knowledge-distillation licensing boundary and label-noise angle. Let me load the WebSearch tool and run the query.

3. assistant (2026-06-16T03:17:54.555Z)

I'll run the search focused on the licensing/governance boundary and label-noise angle.

4. user (2026-06-16T03:18:02.896Z)

Web search results for query: "terms of service prohibit using model outputs to train competing model distillation legal LLM API Gemini OpenAI"

Links: [{"title":"Dispute over AI model distillation tech in OpenAI-DeepSeek case | China","url":"https://law.asia/openai-deepseek-ai-distillation/"},{"title":"How AI could eat itself: Using LLMs to distill rivals","url":"https://www.theregister.com/software/2026/02/14/how-ai-could-eat-itself-using-llms-to-distill-rivals/4567856"},{"title":"OpenAI Terms of service forbid training competitor models via their ML outputs (... | Hacker News","url":"https://news.ycombinator.com/item?id=35375670"},{"title":"The Fine Print of AI Terms of Service: What Most Businesses Miss - TermsFeed","url":"https://www.termsfeed.com/blog/ai-terms-service-fine-print/"},{"title":"How much does distillation really matter for Chinese LLMs?","url":"https://www.interconnects.ai/p/how-much-does-distillation-really"},{"title":"AI Model Distillation Attacks: What They Are and Why They Matter | MindStudio","url":"https://www.mindstudio.ai/blog/ai-model-distillation-attacks-explained"},{"title":"Terms of Use | OpenAI","url":"https://openai.com/policies/row-terms-of-use/"},{"title":"OpenAI says DeepSeek may have \"inappropriately\" used its models' output","url":"https://www.axios.com/2025/01/29/openai-deepseek-ai-models-data-training"},{"title":"OpenAI vs. DeepSeek (Part 2): Fair Use or Foul Play? The AI Training Controversy","url":"https://bitsandbars.substack.com/p/openai-vs-deepseek-part-2-fair-use"},{"title":"Tip Sheets","url":"https://news.cornell.edu/node/330188"}]

Based on the search results, here's what I found about terms of service prohibitions on using model outputs to train competing models:

Terms of Service Restrictions

Companies such as OpenAI, Anthropic, Mistral and xAI include strict "anti-competitive distillation" clauses in their terms of use, prohibiting users from using their services or outputs to develop competing models.

Specifically:

- **OpenAI**: Their terms of use prohibit using Output to develop models that compete with OpenAI. More broadly, you are not allowed to use ChatGPT to violate anyone's rights, modify, copy or lease the services, reverse engineer the model, programmatically extract output, interfere with the services, or create competing models.

- **Google Gemini**: Distillation from Gemini models without permission violates Google's terms of service, and Google can block accounts that do this, or even take users to court.

Background on Model Distillation

Distillation is a common technique developers use to train smaller AI models to replicate the performance of larger, more complex models by training the smaller models on the output of the larger models. However, OpenAI's terms of use forbid users from using outputs from its models to develop other AI models that compete with the company's products and services.

The legality and enforceability of these clauses remains a topic of debate, with some arguing about the intersection with fair use principles.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-16T03:18:05.199Z)

Web search results for query: "knowledge distillation teacher student license terms training on LLM outputs prohibited learning with noisy labels weak supervision distillation 2023 2024"

Links: [{"title":"teacher knowledge distillation for llms","url":"https://www.arxiv.org/pdf/2602.01064"},{"title":"Membership and Memorization in LLM Knowledge Distillation","url":"https://arxiv.org/pdf/2508.07054"},{"title":"Can LLM Watermarks Robustly Prevent Unauthorized Knowledge Distillation?","url":"https://arxiv.org/pdf/2502.11598"},{"title":"Protecting Language Models Against Unauthorized Distillation through Trace Rewriting","url":"https://arxiv.org/pdf/2602.15143"},{"title":"Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models","url":"https://siyan-zhao.github.io/assets/img/opsd/opsd_v3.pdf"},{"title":"Model Distillation: Teacher-Student

Training Guide 2026 | Label Your Data","url":"https://labelyourdata.com/articles/machine-learning/model-distillation"},{"title":"CDLM: Consistency Diffusion Language Models For Faster Sampling","url":"https://arxiv.org/pdf/2511.19269"},{"title":"Knowledge distillation and dataset distillation of large language models: emerging trends, challenges, and future directions - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC12634706/"},{"title":"1 Introduction","url":"https://arxiv.org/html/2408.07238v1"}]

I've completed a web search for your query. Here are the key findings related to knowledge distillation in language models with attention to license terms and training restrictions:

License Terms and Restrictions on Training with LLM Outputs

OpenAI prohibits using output to develop competing models, and Anthropic similarly restricts using their services to develop competing products or train algorithms. Meta's Llama 2 license explicitly states that users cannot use Llama materials or outputs to improve other large language models.

Knowledge Distillation Approaches (2023-2024)

The search results reveal several key developments in LLM knowledge distillation:

Knowledge distillation transfers knowledge from teacher models to student models, and with LLMs treated as black-box models, the rationales generated by the teacher LLM are typically regarded as the embodiment of knowledge.

Various knowledge distillation techniques have been proposed to address privacy risks and computation costs. Early work relied on black-box KD where students imitated outputs from closed-source APIs, and with open-source LLMs, white-box distillation has become prevalent, enabling richer supervision from teacher internal representations like logits or embeddings.

Challenges and Protection Mechanisms

Research has examined whether LLM watermarks can prevent unauthorized knowledge distillation, reflecting concerns about unauthorized use of LLM outputs for training competing models.

The search results emphasize the tension between the technical feasibility of knowledge distillation and license restrictions that prohibit training on proprietary LLM outputs.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:18:15.066Z)

Let me run additional targeted searches on the label-noise handling and confident-learning literature, plus the specific Gemini/Google API terms and the "inference vs training data" governance distinction.

7. user (2026-06-16T03:18:24.943Z)

Web search results for query: "Google Gemini API additional terms generative AI prohibited use train machine learning models develop competing developer policy"

Links: [{"title":"Generative AI APIs Additional Terms of Service | Gemini API | Google AI for Developers","url":"https://ai.google.dev/gemini-api/terms-archive/terms_12_20_24"},{"title":"Generative AI Prohibited Use Policy - Gemini Apps Help","url":"https://support.google.com/gemini/answer/16625148?hl=en"},{"title":"New Google AI Studio Terms Ban Consumer Use: The Developer Pivot","url":"https://www.remio.ai/post/new-google-ai-studio-terms-ban-consumer-use-the-developer-pivot"},{"title":"Gemini API Additional Terms of Service | Google AI for Developers","url":"https://ai.google.dev/gemini-api/terms"},{"title":"The Fine Print of AI Terms of Service: What Most Businesses Miss - TermsFeed","url":"https://www.termsfeed.com/blog/ai-terms-service-fine-print/"},{"title":"Generative AI Prohibited Use Policy - Privacy &

Terms","url":"https://policies.google.com/terms/generative-ai/use-policy"},{"title":"Generative AI Additional Terms of Service - Privacy & Terms","url":"https://policies.google.com/terms/generative-ai"},{"title":"Gemini Apps Privacy Hub - Gemini Apps Help","url":"https://support.google.com/gemini/answer/13594961?hl=en"},{"title":"Gemini Enterprise Agent Platform","url":"https://en.wikipedia.org/wiki/Gemini_Enterprise_Agent_Platform"}]

Based on the search results, here are the key findings regarding Google Gemini API terms and policies on generative AI use:

Competing Models Prohibition

You may not use the Services to develop models that compete with the Services (e.g., Gemini API or Google AI Studio).

Restrictions on Machine Learning Development

You may not use the Services to develop machine learning models or related technology.

Reverse Engineering and Replication

You also may not attempt to reverse engineer, extract or replicate any component of the Services, including the underlying data or models (e.g., parameter weights).

Data Usage for Training

When you use Unpaid Services, including, for example, Google AI Studio and the unpaid quota on Gemini API, Google uses the content you submit to the Services and any generated responses to provide, improve, and develop Google products and services and machine learning technologies. To help with quality and improve our products, human reviewers may read, annotate, and process your API input and output.

Compliance Requirements

In addition to the "API Prohibitions" section in the API Terms, you must comply with our Prohibited Use Policy, which provides additional details about appropriate conduct when using the Services.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-16T03:18:25.719Z)

Web search results for query: "learning with noisy labels co-teaching confident learning DivideMix robust loss survey 2022 2024"

Links: [{"title":"Review?A Survey of Learning from Noisy Labels - IOPscience","url":"https://iopscience.iop.org/article/10.1149/2754-2726/ac75f5"},{"title":"DIVIDEMIX: LEARNING WITH NOISY LABELS AS SEMI- ...","url":"https://openreview.net/pdf?id=HJgExaVtwr"},{"title":"CoDC: Accurate Learning with Noisy Labels via Disagreement and Consistency - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC10886651/"},{"title":"DivideMix: Learning with Noisy Labels as Semi-supervised ...","url":"https://arxiv.org/pdf/2002.07394"},{"title":"Rethinking the impact of noisy labels in graph classification: A utility   and privacy perspective","url":"https://arxiv.org/pdf/2406.07314"},{"title":"Learning from Noisy Labels with Complementary Loss Functions","url":"https://cdn.aaai.org/ojs/17213/17213-13-20707-1-2-20210518.pdf"},{"title":"A Survey on Deep Learning with Noisy Labels: How to train ...","url":"https://arxiv.org/pdf/2012.03061"},{"title":"Label Noise: Ignorance Is Bliss","url":"https://arxiv.org/pdf/2411.00079"},{"title":"[PDF] DivideMix: Learning with Noisy

Labels as Semi-supervised Learning | Semantic Scholar","url":"https://www.semanticscholar.org/paper/DivideMix:-Learning-with-Noisy-Labels-as-Learning-Li-Socher/ce435482acc0e195be8d8f002b2655b4c7b08be6"}]

I've performed the web search for your query. Here are the key findings:

Survey and Methods Overview

The search results include a 2022 survey on learning from noisy labels that covers several important approaches:

DivideMix: DivideMix divides clean labels and noisy labels by fitting a mixed Gaussian distribution, and uses both clean data and noisy labeled data for training. More specifically, the method trains two networks simultaneously and achieves robustness to noise through dataset co-divide, label co-refinement and co-guessing.

Co-teaching: Co-teaching is a method of co-training using two-branch networks, where the training samples of each network come from clean samples screened by the other network based on a small loss criterion for training.

Robust Loss Functions: The results mention various robust loss approaches used in this area, including loss correction methods and complementary loss functions designed to handle noisy labels more effectively than standard cross-entropy loss.

Key Comparison

Co-teaching+ uses the sample selection method but only uses the selected clean data with smaller loss for training without utilizing the data with larger loss. In contrast, DivideMix improves upon this by treating noisy samples as unlabeled data and using semi-supervised learning techniques.

The search results show active research in this area with papers from 2020-2024, covering sampling-based methods, loss correction approaches, and semi-supervised learning combinations for handling noisy labels.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-16T03:18:29.778Z)

I have strong coverage of the licensing boundary (Gemini terms, OpenAI/Anthropic clauses), the distillation-governance literature, and noisy-label methods (DivideMix, co-teaching, confident learning). Let me verify the authoritative Gemini terms page and grab the confident-learning/Cleanlab angle since it's the most directly applicable noisy-label tooling for a small tabular scorer.

10. user (2026-06-16T03:18:37.534Z)

Gemini API Terms: Competing Models, Data Usage, and Training Restrictions

(1) Developing Competing Models

"You may not use the Services to develop models that compete with the Services (e.g., Gemini API or Google AI Studio)."

Additionally, the terms prohibit attempting to "reverse engineer, extract or replicate any component of the Services, including the underlying data or models (e.g., parameter weights)."

(2) Paid vs. Unpaid Services Data Usage

Unpaid Services: Google uses submitted content and generated responses "to provide, improve, and develop Google products and services and machine learning technologies." Human reviewers may read and annotate your inputs and outputs.

Paid Services: "Google doesn't use your prompts...or responses to improve our products." Data is logged only for compliance, safety, and security purposes.

(3) Using Output as Training Data

The terms explicitly state that for Grounding with Google Search, you "will not...cache, frame, syndicate, resell, analyze, **train on**, or otherwise learn from Grounded Results or Search Suggestions."

Similarly, regarding Google Maps grounding: users cannot "**train on** any Google Maps Data."

Restriction: Output from Gemini API cannot be used to train competing models or your own machine learning systems.

11. user (2026-06-16T03:18:40.697Z)

Web search results for query: "confident learning Cleanlab estimating label errors Northcutt 2021 noisy labels small dataset"

Links: [{"title":"Estimating Uncertainty in Dataset Labels","url":"https://dspace.mit.edu/bitstream/handle/1721.1/142946/12125-Article%20(PDF)-26691-1-10-20210414.pdf?sequence=2&isAllowed=y"},{"title":"classification - cleanlab","url":"https://docs.cleanlab.ai/v2.6.4/cleanlab/classification.html"},{"title":"[1911.00068] Confident Learning: Estimating Uncertainty in Dataset Labels","url":"https://arxiv.org/abs/1911.00068"},{"title":"GitHub - cleanlab/cleanlab: Cleanlab's open-source library is the standard data-centric AI package for data quality and machine learning with messy, real-world data and labels. · GitHub","url":"https://github.com/cleanlab/cleanlab"},{"title":"Are Label Errors Imperative? Is Confident Learning Useful? | Towards Data Science","url":"https://towardsdatascience.com/confident-learning-err-did-you-say-your-data-is-clean-ef2597903328/"},{"title":"Improving Label Error Detection and Elimination with Uncertainty Quantification","url":"https://arxiv.org/pdf/2405.09602"},{"title":"Automated Cleanup of the ImageNet Dataset by Model Consensus,   Explainability and Confident Learning","url":"https://arxiv.org/pdf/2103.16324"},{"title":"Con?dent Learning: Estimating Uncertainty in Dataset Labels Curtis G. Northcutt","url":"https://arxiv.org/pdf/1911.00068"},{"title":"PERVASIVE LABEL ERRORS IN TEST SETS ...","url":"https://weasul.github.io/papers/27.pdf"},{"title":"An Introduction to Confident Learning: Finding and Learning with Label Errors in Datasets","url":"https://l7.curtisnorthcutt.com/confident-learning"}]

I've found comprehensive information about confident learning and the Cleanlab work by Northcutt et al. Here's what the search results reveal:

Key Findings

Northcutt, Jiang, and Chuang published "Confident Learning: Estimating Uncertainty in Dataset Labels" in the Journal of Artificial Intelligence Research in 2021.

What is Confident Learning?

Confident learning (CL) is an approach which focuses on label quality by characterizing and identifying label errors in datasets, based on the principles of pruning noisy data, counting with probabilistic thresholds to estimate noise, and ranking examples to train with confidence.

Key Characteristics

CL is a model-agnostic approach for characterizing, identifying, and learning with label errors, which makes use of out-of-sample predicted probabilities to estimate the joint distribution between noisy and correct labels.

The paper shows CL performance exceeding competitive approaches for learning with noisy labels on the CIFAR dataset, and the CL framework is not coupled to a specific data modality or model, with applications to MNIST, Amazon Reviews, and ImageNet.

Practical Implementation

Confident learning only requires predicted probabilities and noisy labels (any model can be used) and does not require any true (guaranteed uncorrupted) labels.

Confident learning is the state-of-the-art for weak supervision, finding label issues in datasets, learning with noisy labels, and uncertainty estimation. The work is available as the open-source Cleanlab package.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-16T03:18:45.169Z)

I have enough to rank the strongest results. The most relevant to the original question's Dimension-3 licensing crux and Dimension-4 label-noise are: the authoritative Gemini API terms, the OpenAI ToS, the confident-learning/Cleanlab work, DivideMix, the co-teaching/noisy-label survey, and the DeepSeek distillation governance case as a real-world enforcement signal.

13. user (2026-06-16T03:19:09.719Z)

Structured output provided successfully

14. assistant (2026-06-16T03:19:22.171Z)

Returned 6 ranked results via StructuredOutput.

Summary of what I found on the **knowledge-distillation licensing boundary + label-noise** angle:

Licensing crux (Dimension 3 governance):
- The authoritative `ai.google.dev/gemini-api/terms` confirms the project's hard constraint is written into Google's own terms ? two distinct prohibitions ("develop models that compete" AND "develop machine learning models or related technology"), plus a reverse-engineer/replicate-weights ban. The load-bearing nuance for the project: the **paid vs. unpaid split** ? paid Gemini does NOT feed your data into Google's ML; unpaid (AI Studio / free quota) does. The clean posture is: paid Gemini as an inference/eval oracle only, never as a training-data source for owned weights.
- OpenAI's ToS (HN thread) and the OpenAI?DeepSeek dispute (law.asia) corroborate this is an industry-standard, actively-enforced clause across all paid teachers ? and flag the unsettled fair-use/enforceability question as unverified legal territory.

Label-noise handling (Dimension 4):
- **Confident Learning / Cleanlab** (Northcutt et al., JAIR 2021) is the highest-leverage, lowest-effort fit: model-agnostic, needs only out-of-sample predicted probabilities + noisy labels, and prunes/ranks suspect labels ? directly usable to triage the tiny golden set or any weak/heuristic pseudo-labels before they reach the numpy logistic scorer. **License caveat to verify:** Cleanlab's package licensing (AGPL concern) should be checked against the MIT/Apache/BSD gate.
- DivideMix and the Song et al. noisy-label survey provide the robust-training toolbox (co-divide/co-teaching, loss correction, semi-supervised treatment of uncertain labels), but these are deep-net methods tuned to large image benchmarks ? adapting them to a 1-D trajectory scorer at n?6?16 is a research bet, not a drop-in.

One honest flag: the top arXiv hit from the first search ("teacher knowledge distillation for llms", arxiv.org/pdf/2602.01064) and a couple of 2602.x IDs look like post-cutoff or unverifiable identifiers, so I excluded them rather than cite something I couldn't confirm.